Muhammad Noor Ul Zaman

Data Scientist | Machine Learning Engineer | Computer Vision Engineer
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Professional Summary

Data Scientist with a Bachelor of Science (B.Sc.) in Data Science and one year of professional experience at ZeroBS (Germany, remote). Built Long Short-Term Memory (LSTM) models for time series anomaly detection and delivered a Flask admin dashboard that improved data readability by 70%. Developed VISIONARY, an end-to-end AI smart city surveillance platform that integrates multiple deep learning models (YOLO, LSTM) into one real-time pipeline with over 95% reported performance. Built Large Language Model (LLM) applications, including Retrieval-Augmented Generation (RAG) and AI agents, and real-time object detection systems. Experienced in the full machine learning lifecycle: data preprocessing, model training and tuning, backend services, and deployment as web applications.

Technical Skills

Programming Languages: Python, R, SQL, Java

Machine Learning and Deep Learning: Machine Learning, Deep Learning, TensorFlow, PyTorch, Keras, LSTM, Time Series Analysis, Time Series Anomaly Detection, Regression, Model Training, Model Optimisation, Hyperparameter Tuning

Computer Vision: Computer Vision, Object Detection, OpenCV, YOLO, YOLOv5, YOLOv8, Real-Time Video Analytics

LLM and Generative AI: Large Language Models (LLMs), Generative AI, Retrieval-Augmented Generation (RAG), LangChain, OpenAI API, GPT-4o, FAISS, Pinecone, ChromaDB, Vector Search, Embeddings, AI Agents

Data Analysis: Pandas, NumPy, Seaborn, Data Cleaning, Data Preprocessing, Natural Language Processing (NLP), Sentiment Analysis, SPSS, SAS, Excel, Google Analytics

Data Visualisation: Tableau, Power BI

Backend and Deployment: Flask, FastAPI, Streamlit, Docker, Redis, REST API, Domain-Driven Design (DDD), Automation

Tools: Git, GitHub, Selenium, Version Control, DevOps

Work Experience

Data Scientist - ZeroBS

Kiel, Germany (Remote) | January 2025 - December 2025

- Developed machine learning models for time series anomaly detection using LSTM, in collaboration with the client.
- Built and deployed an interactive Flask admin dashboard that improved data readability and understanding by 70% and streamlined system management.
- Applied Domain-Driven Design (DDD) architecture to improve code readability, simplify debugging, and strengthen error handling.
- Supported DevOps tasks, version control, and collaborative development using GitHub.
- Supported Python development including automation, data preprocessing, and backend services.
- Collaborated with cross-functional teams to deliver scalable and maintainable solutions.

Projects

Side projects built alongside my role at ZeroBS (December 2024 - December 2025).

StaffViz: Autonomous HR AI Agent

September 2025 - December 2025 | Technologies: OpenAI GPT-4o, LangChain, ChromaDB, Selenium, FastAPI

- Built an agentic HR assistant (team project with a colleague) capable of autonomous web browsing, answering policy queries, and managing leave workflows using LangChain tool use.
- Reduced HR ticket resolution time by 60% at a company of 200+ employees.

Intelligent ATS: AI-Powered Resume Screening Engine
July 2025 - August 2025 | Technologies: OpenAI, FAISS, LangChain, FastAPI, Pandas
- Built an enterprise applicant tracking system using OpenAI embeddings and vector search to score, rank, and explain candidate-job fit with explainable AI output.
- Reduced recruiter screening workload by 70% through structured ranking reports.

Enterprise RAG System: Multi-Source Knowledge Engine
April 2025 - June 2025 | Technologies: LangChain, Pinecone, FAISS, OpenAI, FastAPI, Redis
- Built a production RAG pipeline ingesting PDFs, databases, and APIs into a hybrid vector and keyword search system.
- Implemented re-ranking, query expansion, and contextual compression, achieving 92% answer accuracy.

Smart Security System: Real-Time Weapon Detection
December 2024 - January 2025 | Technologies: YOLOv8, OpenCV, FastAPI, Python, Docker
- Developed a standalone real-time AI weapon detection system using YOLOv8, deployed via FastAPI with an alert notification pipeline (derived from the weapon detection work in VISIONARY).
- Achieved 94% mAP on a custom dataset designed for high-risk environment deployment.

VISIONARY: AI-Powered Smart City Surveillance System
September 2023 - August 2024 | Technologies: YOLO, LSTM, OpenCV, Flask, Python
- Built an end-to-end AI surveillance platform that detects, analyses, and reports urban safety and infrastructure incidents from video footage, achieving over 95% reported performance.
- Developed YOLO-based detection modules for fire and smoke, potholes, wrong-way vehicles, waste, and violence, with automated security alerts.
- Implemented an LSTM-based temporal analysis module to detect accidents from sequential video patterns that single frames cannot reliably show.
- Improved individual models through dataset preparation, training, evaluation, tuning, and performance optimisation.
- Integrated multiple specialised models into a single real-time pipeline that processes footage, detects events, and generates alerts.
- Built a Flask monitoring dashboard giving a central view of AI predictions, detected incidents, and surveillance information.

Age and Gender Detection System
May 2023 | Technologies: PyTorch, Streamlit, Python
- Developed a PyTorch deep learning model for age and gender prediction from facial images, trained on a Kaggle dataset.
- Deployed with Streamlit as an interactive web app where users upload images and receive instant predictions.

Sentiment Analysis on Tweets
February 2023 | Technologies: R, RStudio, Regression, Machine Learning
- Ran sentiment analysis on 14,000 tweets reviewing six major airlines; cleaned and analysed text data to identify the main complaints for each airline.
- Applied regression and machine learning to produce a final model with 88% accuracy and 34% lift.

Education

Bachelor of Science (B.Sc.) in Data Science - GIFT University
Gujranwala, Pakistan | November 2020 - September 2024

Languages

English: Advanced (C1 listening and reading, B2 writing and speaking)
Urdu: Native